

1.INTRODUCTION

This chapter contains a hybrid text summarization, the problem statement and problem description, project objectives, advantages and challenges, applications and the organization of the report.

1.1 Heart Attack Prediction.

A heart attack, medically known as myocardial infarction, is a serious cardiovascular condition that occurs when the blood supply to a part of the heart muscle is blocked or significantly reduced, preventing oxygen and essential nutrients from reaching the heart tissue. This blockage is most commonly caused by the buildup of fatty deposits, cholesterol, and other substances in the coronary arteries, a condition known as atherosclerosis. When these deposits rupture, they can form a blood clot that obstructs blood flow, leading to damage or death of heart muscle cells. Several risk factors contribute to the occurrence of heart attacks, including high blood pressure, high cholesterol levels, diabetes, smoking, obesity, physical inactivity, stress, and a family history of heart disease. The heart plays a vital role in the human body by continuously pumping oxygen-rich blood to organs and tissues, supplying nutrients, and removing waste products. Since all body systems depend on the efficient functioning of the heart, any disruption in its blood supply can have severe consequences. Therefore, understanding the causes and risk factors associated with heart attacks is essential for early diagnosis, prevention, and effective treatment, ultimately helping to reduce mortality and improve overall cardiovascular health.

With the rapid growth of digital technologies, large amounts of healthcare data are being generated and stored every day. This has created opportunities to use advanced analytical techniques to extract meaningful information from medical records and support healthcare decision-making. Machine learning has emerged as one of the most promising technologies in this field because of its ability to learn from historical data and identify patterns that are often difficult to detect through traditional methods. By analyzing patient information, machine learning models can assist in estimating the likelihood of future health conditions and provide valuable support for preventive healthcare.

The process of developing an effective heart attack prediction system begins with the collection of patient-related data containing various medical and demographic characteristics. These attributes may include age, gender, cholesterol level, blood pressure, heart rate, chest pain type, blood sugar level, and other factors that are commonly associated with cardiovascular health. However, healthcare datasets are rarely perfect.

They often contain missing values, duplicate entries, inconsistencies, and unusual observations that can affect the performance of machine learning models. For this reason, the first step involves carefully examining and preparing the data before it is used for analysis. Data cleaning plays an essential role in improving the overall quality of the dataset. Missing values are handled appropriately, duplicate records are removed, and inconsistencies are corrected to ensure that the information used for training is accurate and reliable. Outliers are identified and analyzed to determine whether they represent genuine medical conditions or abnormal observations that could negatively affect the learning process. By preparing the dataset in a structured and meaningful manner, the foundation is established for building reliable predictive models.

Once the data has been cleaned and organized, exploratory data analysis is performed to gain a deeper understanding of the dataset. This stage helps reveal important relationships between different variables and provides insights into how various health factors may influence the risk of heart disease. Visualizations, statistical summaries, and correlation analyses make it easier to identify trends, patterns, and potential risk indicators within the data. Understanding these relationships is important because it allows meaningful features to be selected and helps improve the effectiveness of the prediction models.

After exploring the dataset, several machine learning algorithms are applied and evaluated. Different algorithms approach the prediction task in different ways, and each has its own strengths when dealing with healthcare data. Logistic Regression provides a simple yet effective approach for estimating the probability of heart disease occurrence. Decision Trees create a series of logical decisions that make the prediction process easier to understand. Random Forest improves prediction reliability by combining the outputs of multiple decision trees and reducing the possibility of overfitting. K-Nearest Neighbors identifies patterns based on similarities between patients, while Support Vector Machine focuses on finding the most

effective boundary between different classes. Ensemble techniques such as AdaBoost and Gradient Boosting further improve predictive performance by combining multiple models to produce stronger results. XGBoost, which is an advanced boosting algorithm, is widely recognized for its efficiency, speed, and ability to achieve high prediction accuracy on complex datasets.

Many existing heart attack prediction systems primarily focus on generating prediction results without addressing some of the common challenges associated with machine learning models. One such challenge is overfitting, where a model performs exceptionally well on training data but fails to maintain the same level of accuracy when presented with new and unseen data. This can reduce the practical usefulness of the system in real-world healthcare environments. To overcome these limitations, careful preprocessing, feature selection, model tuning, and performance evaluation techniques are applied throughout the development process. Comparing multiple algorithms and selecting the most suitable approach helps improve overall accuracy and ensures that the system produces more reliable predictions.

Although obtaining accurate predictions is important, providing explanations for those predictions is equally valuable, especially in healthcare applications. Patients and healthcare professionals often need to understand why a particular prediction was made rather than simply receiving a result. Traditional machine learning models sometimes act like black boxes, making decisions without clearly explaining the factors that influenced the outcome. This lack of transparency can make users hesitant to trust the system.

To address this issue, Explainable Artificial Intelligence (XAI) techniques are incorporated into the prediction framework. Instead of only displaying whether a person has a low or high risk of a heart attack, the system also explains how individual features contributed to the prediction. For example, it can show whether factors such as age, cholesterol level, blood pressure, chest pain type, or heart rate had a positive or negative influence on the final result. This additional layer of explanation makes the prediction process more transparent and helps users understand the reasoning behind the outcome.

Another valuable aspect of the system is its ability to maintain records of newly entered patient information. Whenever a user provides health details for prediction, the data is stored in a separate document for future reference and analysis. Maintaining these records creates opportunities for continuous monitoring, future research, and further improvement of the prediction models. As more patient data becomes available over time, the stored information can contribute to enhancing model performance and expanding the system's knowledge base.

The integration of data preprocessing, exploratory analysis, multiple machine learning algorithms, model optimization techniques, and explainable artificial intelligence creates a comprehensive framework for heart attack risk assessment. Rather than functioning solely as a prediction tool, the system provides meaningful insights into the factors influencing cardiovascular risk while delivering improved predictive performance. In an era where healthcare is increasingly driven by data and technology, such systems have the potential to support early risk detection, promote preventive healthcare practices, and assist both healthcare professionals and individuals in making informed decisions about heart health. By combining predictive capability with transparency and continuous data management, the proposed approach represents a practical and modern solution for addressing one of the most significant healthcare challenges of the present day.

1.2 Problem Description and Statement

1.2.1 Problem Description

The core problem addressed in this study is the accurate and reliable prediction of heart attack risk using patient health data. Cardiovascular diseases remain a major cause of mortality worldwide, making early risk assessment essential for timely medical intervention and preventive care. Traditional methods of risk evaluation often rely on manual analysis and clinical expertise, which can be time-consuming and may not efficiently utilize the growing volume of healthcare data.

machine learning techniques have demonstrated significant potential in predicting heart attack risk, many existing models primarily focus on prediction outcomes without providing sufficient insight into the factors influencing those predictions. This lack of interpretability can reduce user confidence and limit the practical adoption of such systems in healthcare settings. Furthermore, achieving high predictive performance while maintaining transparency and consistency remains a significant challenge.

This study focuses on developing a heart attack risk prediction model that combines advanced machine learning algorithms with Explainable Artificial Intelligence (XAI) techniques. By incorporating data preprocessing, exploratory data analysis, feature selection, and model optimization strategies, the proposed approach aims to improve prediction performance while providing clear explanations of feature contributions. The integration of predictive analytics and interpretability is expected to enhance the reliability, usability, and effectiveness of the system, thereby supporting informed healthcare decision-making and promoting early detection of cardiovascular risks.

1.2.2 Problem statement.

Heart attacks are among the leading causes of mortality worldwide, highlighting the need for accurate and timely risk assessment. Existing machine learning-based prediction models can identify individuals at risk; however, many of these models function as black-box systems, providing little insight into how predictions are made. This lack of interpretability reduces transparency and limits the confidence of healthcare professionals in relying on the results. Furthermore, variations in model performance and data utilization can affect prediction reliability. Therefore, there is a need for an intelligent heart attack risk prediction system that not only achieves high predictive accuracy but also explains the contribution of individual health factors to the prediction outcome. Such a system can support early detection, enhance clinical decision-making, and improve preventive healthcare strategies.

1.3 Objectives

Following are the Objectives of the project undertaken:

- To predict the probability of suffering of any patient from heart attacks in the upcoming future. Identifying the major factors influencing heart attacks.
- Identifying cholesterol level with higher possibilities of heart attacks.
- Identifying chest pain types having higher possibilities of heart attacks.
- Recommendations for preventing / reducing chances of getting a heart attacks.
- To integrate Explainable AI techniques to understand how different features contribute to the model's prediction.

1.4 Advantages of this project

The advantages of the hybrid text summarization using deep learning techniques are listed as follows:

1. Early Detection of Heart Attack Risk

The proposed system helps identify individuals who may be at risk of experiencing a heart attack before serious complications arise. By examining important health-related factors such as age, blood pressure, cholesterol levels, diabetes status, and physical activity, the model can recognize warning signs at an early stage. This enables both patients and healthcare professionals to take appropriate preventive measures, improving overall heart health and reducing the likelihood of severe cardiovascular events.

2. Accurate and Dependable Risk Prediction

The system applies machine learning techniques to learn patterns from patient health records and generate reliable risk predictions. By analyzing multiple factors simultaneously, it can uncover relationships that may be difficult to identify through conventional assessment methods. This capability enhances the consistency and accuracy of risk evaluation, supporting healthcare professionals in making better-informed clinical decisions.

3. Clear and Understandable Prediction Results

A key strength of the system is its ability to explain how predictions are made. Through the integration of Explainable Artificial Intelligence (XAI) methods such as SHAP, users can view the influence of each health factor on the final prediction. This transparency helps healthcare providers better interpret the results and allows patients to gain a clearer understanding of the factors affecting their cardiovascular health.

4. Individualized Patient Evaluation

Every patient has a unique combination of health conditions and lifestyle characteristics. Instead of relying on a one-size-fits-all approach, the system evaluates each individual based on their specific health information. This personalized assessment provides more relevant risk estimates and supports the development of targeted prevention strategies, treatment recommendations, and lifestyle modifications.

5. Rapid Support for Medical Decision-Making

The system is capable of generating heart attack risk predictions within a short period after patient information is entered. This reduces the time required for manual analysis and allows healthcare professionals to quickly assess a patient's condition. Faster access to risk information can improve clinical efficiency, assist in prioritizing patients who require immediate attention, and support timely medical intervention when necessary.

1.5 Limitations of this project

The limitations of the Heart attack prediction using machine learning are listed as follows:

1. Dependence on Data Quality

The effectiveness of the proposed system relies on the quality of the data used during model development. If the dataset contains incomplete records, incorrect values, or inconsistencies, the model's ability to learn meaningful patterns may be affected. **Limited Availability of Risk Factors.** The prediction model evaluates heart attack risk using the health attributes available in the dataset. However, certain factors that can influence cardiovascular health, such as family medical history, stress levels, dietary habits, medication use, and other detailed clinical information, are not included.

2. Restricted Feature Set

One of the main limitations of this project is that the prediction model relies only on the clinical and demographic attributes available in the dataset. Features such as age, cholesterol level, blood pressure, heart rate, and other recorded health parameters are used to estimate the risk of a heart attack. However, heart disease is influenced by a wide range of factors that may not be included in the dataset.

3. Dependence on User Input

The accuracy of the generated prediction depends on the information provided by the user. If the entered health details are incomplete, inaccurate, or do not reflect the individual's actual condition, the prediction outcome may not accurately represent the person's risk level. Therefore, accurate data entry is essential for obtaining meaningful results.

4. Understanding Prediction Explanations

The system provides explanations for its predictions through Explainable Artificial Intelligence techniques, helping users understand the factors that influence the outcome. However, some users may find it challenging to interpret feature contribution scores without basic knowledge of the explanation process.

1.6 Applications of this project

The applications of Heart attack prediction using machine learning are as follows:

1. Hospitals and Healthcare Centers

The proposed heart attack prediction system can support healthcare professionals in identifying patients who are at a higher risk of cardiovascular complications. By analyzing patient health data, the system provides quick and reliable risk assessments that assist doctors in diagnosis, patient monitoring, and treatment planning. Early identification of high-risk individuals enables timely medical intervention and improves the quality of patient care.

2. Telemedicine and Remote Healthcare Services

The system can be integrated into telemedicine platforms to provide remote cardiovascular risk assessment. Patients can enter their health information and receive an initial evaluation without visiting a healthcare facility. This is particularly beneficial for individuals living in remote areas or those with limited access to specialized medical services. Healthcare providers can use the prediction results and explanatory insights during virtual consultations to support informed medical decisions and preventive care.

3. Medical Education and Clinical Training

The system serves as a valuable learning tool for medical students and healthcare professionals. Through Explainable Artificial Intelligence (XAI), users can understand how different risk factors contribute to heart attack prediction. This enhances knowledge of cardiovascular risk assessment, strengthens clinical reasoning skills, and demonstrates the practical application of machine learning in healthcare.

4. Integration with Smart Healthcare Systems

The prediction model can be incorporated into smart healthcare infrastructures and electronic health record systems. Automated analysis of patient information enables continuous risk monitoring and supports data-driven clinical decision-making. This integration improves healthcare efficiency, reduces manual workload, and facilitates proactive patient management.

5. Rural Healthcare Services

In rural and underserved regions where access to cardiologists and advanced healthcare facilities may be limited, the system can function as an effective preliminary screening tool. Healthcare workers can use it to identify individuals who may require further medical evaluation and referral to specialized centers, thereby improving access to quality cardiovascular care.

6. Routine Health Screening Programs

The system can be utilized during community health screening camps and public health initiatives to assess cardiovascular risk among large populations. By quickly identifying individuals with elevated risk levels, healthcare professionals can recommend preventive measures, lifestyle modifications, and further diagnostic testing, contributing to improved public health outcomes.

7. Wearable Health Monitoring Systems

The model can be integrated with wearable devices such as smartwatches and fitness trackers that collect real-time health information. Continuous monitoring of health parameters combined with predictive analytics enables early detection of potential cardiovascular risks. The system can alert users when abnormal patterns are observed, encouraging timely medical consultation and proactive health management.

1.7 Organization of the report.

The information presented in this report is organized into five chapters. **Chapter 1** introduces the Heart Attack Prediction System using Machine Learning and Explainable Artificial Intelligence (XAI). It discusses the background of the study, problem statement, objectives, scope, advantages, limitations, applications, and report organization. **Chapter 2**

presents a comprehensive review of existing literature related to heart disease prediction, machine learning algorithms..**Chapter 3** describes the proposed system and methodology adopted for the study. It includes data collection, data preprocessing, exploratory data analysis, model development, performance evaluation. **Chapter 4** presents the experimental results, comparative analysis of machine learning models, prediction outcomes, and feature contribution analysis using SHAP, along with different test cases used for system validation. **Chapter 5** summarizes the overall findings of the study, highlights the contributions of the proposed system, and discusses potential future enhancements for improving prediction performance and usability.

1.8 Summary

This chapter introduced the proposed **Heart Attack Prediction System based on Machine Learning and Explainable Artificial Intelligence (XAI)** and discussed its importance in supporting the early identification of cardiovascular risk. The growing impact of heart-related diseases and the need for accurate risk assessment methods were highlighted to establish the motivation behind the development of the system.

Section 1.1 provided an overview of heart attack prediction and explained how machine learning techniques can be used to analyze patient health data and estimate risk levels. **Section 1.2** described the problem addressed by the project and outlined the challenges associated with traditional risk assessment approaches. The goals and expected outcomes of the proposed system were presented in **Section 1.3**.

The benefits offered by the system, including improved prediction capability and interpretability of results, were discussed in **Section 1.4**. **Section 1.5** examined the limitations of the proposed approach and identified factors that may influence its performance. The practical uses of the system in healthcare, education, remote patient monitoring, and other real-world environments were explained in **Section 1.6**. Finally, **Section 1.7** outlined the structure of the report and provided a brief overview of the topics covered in the remaining chapters.

1. LITERATURE REVIEW

The literature review is an important part of this project because it examines previous research related to heart attack prediction, machine learning, and Explainable Artificial Intelligence (XAI) in healthcare. Its main purpose is to understand the existing work in the field and identify suitable methods for developing the proposed system. By studying earlier research, it becomes possible to learn about different prediction techniques, their advantages, and their limitations. It also helps in identifying research gaps and areas where improvements can be made. In addition, the literature review supports the selection of appropriate machine learning algorithms, data processing methods, and evaluation techniques for the project. Overall, it provides the necessary background for the study, highlights the importance of the research, and serves as a foundation for the development of the proposed heart attack prediction system.

2.1 Literature Survey:

The following section presents a review of selected research studies related to **heart attack prediction, machine learning techniques in healthcare**. Each study is briefly discussed with reference to the respective authors and their contributions to the field. This review provides an overview of the various approaches, methodologies, and prediction models that have been developed for assessing cardiovascular risk. It also highlights the advancements made in heart disease prediction over time and helps establish the foundation for the proposed heart attack prediction system.

The study by **Ranjit Shrestha and Jyotir Moy Chatterjee [1]** proposed a web-based heart disease prediction system using machine learning techniques. The primary objective of the study was to support the early prediction of heart disease, which remains one of the leading causes of mortality worldwide. The authors developed a prediction system by applying **Decision Tree (DT)** and **Naïve Bayes (NB)** classification algorithms and compared their performance in predicting heart disease risk. The **UCI Heart Disease dataset** was used to train and evaluate the prediction models. The study highlighted the role of machine learning in analyzing patient health data and identifying patterns associated with heart disease. The results demonstrated that both algorithms were effective in predicting heart disease, making the system a useful tool for supporting healthcare professionals in early diagnosis and risk assessment.

The research emphasized the potential of machine learning-based prediction systems in improving healthcare services and assisting in preventive medical decision-making.

The study by **Ochin Sharma [2]** investigated the use of various machine learning algorithms for heart attack prediction and analysis. The research highlighted that healthcare systems generate large amounts of patient data, but much of this information is not effectively utilized for decision-making. To overcome this challenge, the study applied data mining and machine learning techniques to identify patterns associated with heart disease. Several classification methods, including decision trees, Bayesian networks, rule-based approaches, and artificial neural networks, were examined for their effectiveness in predicting heart attacks. The main objective of the study was to determine the most suitable algorithm for accurate heart attack prediction. The results showed that machine learning techniques can successfully analyze healthcare data and support the early detection of heart attack risk, helping healthcare professionals make better clinical decisions.

The study by **Dhai Eddine Salhi, Abdelkamel Tari, and M.-Tahar Kechadi [3]** explored the use of machine learning techniques for heart disease prediction from a data analytics perspective. The research focused on identifying patients at risk of heart disease by analyzing healthcare data. Initially, a data preprocessing stage was performed, where the most relevant features were selected using a correlation matrix to improve model performance. The study then applied three machine learning algorithms, namely **Neural Networks, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)**, on datasets of different sizes to evaluate their accuracy and stability. The experimental results showed that the Neural Network model achieved the best performance, reaching an accuracy of approximately **93%**. The study concluded that machine learning techniques, particularly Neural Networks, can be effectively used for accurate heart disease prediction and can support healthcare professionals in early risk assessment and decision-making.

The study by **Najwan Alshraideh, Abedalrahman Alshraideh, Mohammad Alshraideh, Yara Alkayed, Yasmin Al Trabsheh, and Bahaaldeh Alshraideh [4]** focused on improving heart disease prediction using machine learning techniques on data collected from Jordan University Hospital. The research applied several algorithms, including **Random Forest, Support Vector Machine (SVM), Decision Tree, Naïve Bayes, and K-Nearest Neighbors**

(KNN), along with **Particle Swarm Optimization (PSO)** for feature selection. By selecting the most relevant features from the dataset, the proposed model achieved a prediction accuracy of **94.3%**. Among the evaluated methods, the **SVM classifier combined with PSO** produced the best results. The study demonstrated that machine learning and feature optimization techniques can significantly improve the accuracy of heart disease prediction and support early diagnosis and effective clinical decision-making.

The study by **Suraj Kumar Gupta, Aditya Shrivastava, S. P. Upadhyay, and Pawan Kumar Chaurasia [5]** investigated the use of machine learning techniques for heart attack prediction. The research applied several supervised learning algorithms, including **Gradient Boosting, Decision Tree, Random Forest, and Logistic Regression**, using data from the **Framingham and UCI Heart Disease** datasets. The primary objective was to predict the likelihood of heart attacks and identify the most influential risk factors. The results showed that the **Gradient Boosting classifier** achieved the highest prediction accuracy. The study also identified factors such as chest pain type, cholesterol level, heart rate, and age as significant contributors to heart attack risk, demonstrating the effectiveness of machine learning in supporting early disease prediction and preventive healthcare.

The study by **Neha Nandal, Lipika Goel, and Rohit Tanwar [6]** proposed a machine learning-based heart attack prediction method aimed at improving the early detection of cardiovascular diseases. The research analyzed various risk factors such as high blood pressure, high cholesterol, abnormal pulse rate, and diabetes using machine learning techniques. Several algorithms, including **Support Vector Machine (SVM), Logistic Regression, Naïve Bayes, and XGBoost**, were applied to data obtained from the **UCI Machine Learning Repository**. The performance of the models was evaluated using metrics such as accuracy, precision, recall, and area under the curve (AUC). The results showed that **XGBoost achieved the best performance with an AUC of 0.94**, followed by Logistic Regression with an AUC of 0.92. The study demonstrated that machine learning models, particularly boosting algorithms, can effectively support the accurate prediction of heart attack risk and assist in early clinical decision-making.

The study by **Akshit Bhardwaj, Ayush Kundra, Bhavya Gandhi, Sumit Kumar, Arvind Rehalia, and Manoj Gupta [7]** focused on predicting heart attacks using machine learning techniques. The research highlighted the growing impact of cardiovascular diseases and emphasized the importance of early risk prediction in reducing heart-related deaths. The primary objective was to identify the most significant risk factors associated with heart disease and estimate an individual's likelihood of developing the condition. The prediction model was trained using the **Framingham Heart Study dataset**, which contains health and lifestyle information from a large population. By analyzing factors such as lifestyle habits and dietary patterns, the system was able to estimate heart disease risk and support preventive healthcare measures. The study demonstrated that machine learning can be effectively used to assist in the early detection of heart disease and encourage healthier lifestyle choices.

The study by **Annapurna Anant Shanbhag, Chinmai Shetty, Alaka Ananth, Anjali Shridhar Shetty, K. Kavanashree Nayak, and B. R. Rakshitha [8]** focused on analyzing the probability of heart attack risk using machine learning techniques. The research aimed to identify individuals who may be at risk of heart attacks by evaluating important health parameters such as **cholesterol level, blood pressure, and blood sugar**. Various machine learning models were applied to predict heart attack risk, and the results were displayed through a web-based application. In addition, the system provided precautionary measures under the guidance of cardiologists when a potential risk was detected. The study found that the **Support Vector Machine (SVM)** model achieved the highest prediction accuracy of **85.7%**, demonstrating the effectiveness of machine learning in supporting early heart disease risk assessment and preventive healthcare.

The study by **Diksha Bhardwaz, Yashank Singla, Rupinder Kaur, and Liza Sharma [9]** investigated the use of **XGBoost models** for heart attack prediction and cardiovascular risk assessment. The research developed both **XGBRegressor** and **XGBClassifier** models to predict heart attack risk and evaluate disease severity. A **10-fold cross-validation** technique was used to ensure the reliability and robustness of the models. The results showed that the classification model achieved an excellent **ROC-AUC score of 0.9973**, while the regression model produced a low prediction error, demonstrating strong predictive performance. The study highlighted the effectiveness of advanced machine learning algorithms in identifying

individuals at risk of cardiovascular diseases and emphasized their potential for supporting early diagnosis, preventive care, and improved healthcare decision-making

The study by **Brandon N. Nava-Martinez, Sahid S. Hernandez-Hernandez, Denzel A. Rodriguez-Ramirez, Jose L. Martinez-Rodriguez, Ana B. Rios-Alvarado, Alan Diaz-Manriquez, Jose R. Martinez-Angulo, and Tania Y. Guerrero-Melendez [10]** proposed a machine learning framework for real-time heart attack risk prediction and clinical decision support. The framework combined advanced data preprocessing, feature selection, and hyperparameter optimization techniques to improve prediction performance. The researchers evaluated six traditional machine learning models along with several stacking and voting ensemble models using multiple clinical datasets, outperforming many existing methods. The study also developed a lightweight web-based application that enables real-time risk assessment, demonstrating the potential of ensemble learning techniques to support early diagnosis and improve cardiovascular healthcare outcomes.

The study by **Kayalvily A/P Tabianan, Chew Lian Wei, and Deshinta Arrova Dewi [11]** proposed an expert system for heart attack prediction aimed at increasing awareness and supporting the early detection of heart attack risk. The research highlighted that heart attacks are a major cause of death, particularly in low- and middle-income countries where limited financial resources often restrict access to regular healthcare services. To address this issue, the authors developed an **Android-based expert system** that allows users to assess their heart health status through a mobile application. The system utilizes a **decision table** to analyze user inputs and determine the potential risk of heart attacks. By providing an accessible and user-friendly platform, the proposed system enables individuals to identify possible heart-related risks at an early stage and seek timely medical attention. The study demonstrated the potential of mobile health technologies in improving awareness, accessibility, and preventive cardiovascular care.

2.2 Overview of the Literature Survey

Authors (Year)	Dataset Used	Models/Methods/Techniques Applied	Performance / Key Findings
Ranjit Shrestha and Jyotir Moy Chatterjee (2021)	UCI Heart Disease Dataset	Decision Tree (DT), Naïve Bayes (NB)	Both algorithms effectively predicted heart disease risk and supported early diagnosis.
Ochin Sharma (2021)	Healthcare Heart Disease Data	Decision Tree, Bayesian Network, Rule-Based Methods, Artificial Neural Networks	Machine learning techniques successfully identified heart attack risk factors and improved prediction accuracy.
Dhai Eddine Salhi et al. (2022)	Heart Disease Dataset	Correlation-Based Feature Selection, Neural Network, SVM, KNN	Neural Network achieved the highest accuracy of approximately 93%.

Najwan Alshraideh et al. (2022)	Jordan University Hospital Dataset	Random Forest, SVM, Decision Tree, Naïve Bayes, KNN, PSO	SVM with Particle Swarm Optimization achieved 94.3% prediction accuracy.
Suraj Kumar Gupta et al. (2022)	Framingham and UCI Heart Disease Datasets	Gradient Boosting, Decision Tree, Random Forest, Logistic Regression	Gradient Boosting outperformed other models and identified significant risk factors.
Neha Nandal et al. (2023)	UCI Machine Learning Repository	SVM, Logistic Regression, Naïve Bayes, XGBoost	XGBoost achieved the highest performance with an AUC score of 0.94.
Akshit Bhardwaj et al. (2023)	Framingham Heart Study Dataset	Machine Learning-Based Risk Prediction Models	Successfully identified lifestyle and health-related risk factors for heart disease prediction.
Annapurna Anant Shanbhag et al. (2023)	Heart Attack Dataset	SVM and Other ML Models, Web-Based Application	SVM achieved the highest accuracy of 85.7% for heart attack

			risk prediction.
Diksha Bhardwaz et al. (2024)	Cardiovascular Disease Dataset	XGBClassifier, XGBRegressor, 10-Fold Cross Validation	XGBClassifier achieved an ROC-AUC score of 0.9973, demonstrating excellent predictive performance.
Brandon N. Nava-Martinez et al. (2024)	Multiple Clinical Datasets	Ensemble Learning, Stacking, Voting Classifiers, Hyperparameter Optimization	Ensemble models improved prediction accuracy and enabled real-time risk assessment.
Kayalvily A/P Tabianan et al. (2024)	User Health Input Dataset	Android-Based Expert System, Decision Table	Developed a mobile application for early heart attack risk assessment and awareness.

Table 2.1.1 Overview of the Literature Survey

2.3 Summary

This chapter presented a review of existing research related to **heart attack prediction using machine learning and artificial intelligence techniques**. **Section 2.1** discussed various studies that applied different machine learning algorithms, including Decision Tree, Naïve Bayes, SVM, KNN, Random Forest, Logistic Regression, Neural Networks, XGBoost, and Ensemble Learning for cardiovascular risk prediction.

Section 2.2 provided a comparative analysis of the reviewed studies based on factors such as authors, datasets, techniques used, and model performance. The literature review showed that advanced machine learning approaches, especially ensemble and boosting methods, often achieve higher prediction accuracy. It also highlighted the importance of Explainable Artificial Intelligence (XAI) in improving the interpretability of prediction results. Overall, the review helped identify suitable techniques and methodologies that formed the foundation for the development of the proposed Heart Attack Prediction System.

3. METHODOLOGY

This chapter describes the methodology adopted for developing the **Heart Attack Prediction System using Machine Learning and Explainable Artificial Intelligence (XAI)**. It explains the systematic process followed to collect, preprocess, and analyze patient health data for predicting heart attack risk. The methodology focuses on building an accurate and reliable prediction model while providing clear explanations of the factors influencing the prediction results. It also outlines the overall workflow used in the implementation of the proposed system, including data preprocessing, exploratory data analysis, model development, performance evaluation, and the integration of XAI techniques for prediction interpretation.

3.1 Proposed Methodology:

The proposed methodology outlines the process used to develop the **Heart Attack Prediction System using Machine Learning and Explainable Artificial Intelligence (XAI)**. The process begins with data collection and preprocessing to prepare the dataset for analysis. Exploratory Data Analysis (EDA) is performed to understand the data and identify important patterns. Machine learning algorithms are then trained and evaluated to select the most suitable model for heart attack prediction. SHAP-based Explainable AI techniques are integrated to explain how different features influence the prediction results. This methodology helps develop an accurate, reliable, and interpretable system for heart attack risk assessment.

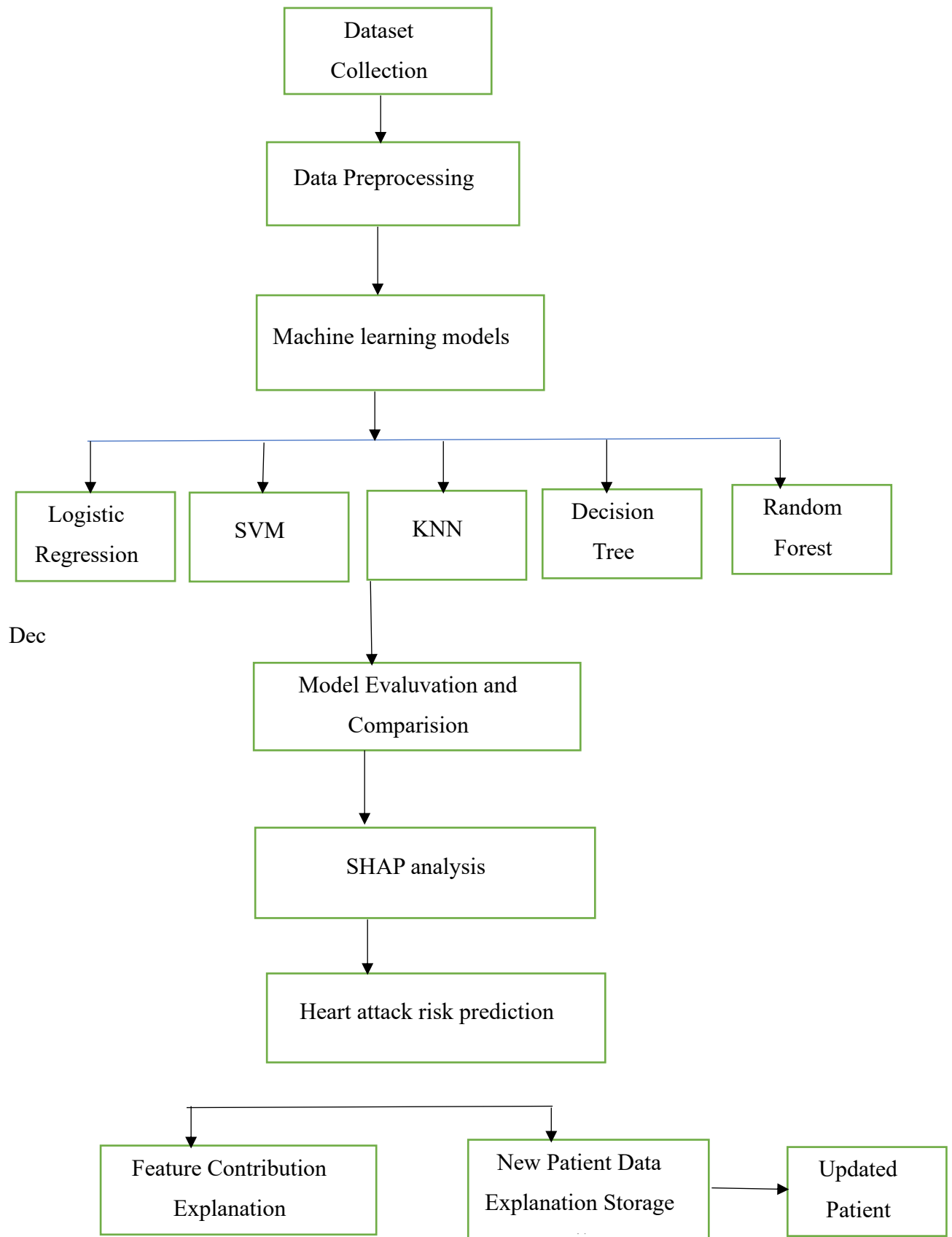


Figure 3.1: Block Diagram for Heart Attack Prediction

Figure 3.1 illustrates the architecture and working process of the proposed Heart Attack Risk Prediction System. The system is developed to predict the risk of heart attacks using machine learning techniques and to provide interpretable explanations for each prediction through SHAP (SHapley Additive Explanations).

The first stage involves collecting the Heart Disease Dataset from the UCI Machine Learning Repository, which is a widely recognized and reliable source for medical and machine learning research. The dataset contains various patient health attributes and cardiovascular risk factors such as age, gender, cholesterol level, blood pressure, and other clinical indicators associated with heart disease. These features serve as input variables for training and evaluating the machine learning models.

After dataset collection, data preprocessing is performed to improve the quality of the data and prepare it for analysis. This stage includes handling missing values, removing duplicate records, checking data consistency, and converting categorical variables into numerical formats suitable for machine learning algorithms. Feature scaling techniques are also applied where necessary to ensure that all attributes contribute effectively during model training. The preprocessing process helps create a clean, accurate, and structured dataset, thereby improving the performance and reliability of the prediction models.

Once the data has been preprocessed, multiple machine learning algorithms are employed to develop predictive models. The algorithms used in this study include Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, and Random Forest. Logistic Regression estimates the probability of heart attack risk based on patient characteristics, while SVM identifies an optimal boundary that separates patients into different risk categories. KNN predicts risk by analyzing similarities between patient records, whereas the Decision Tree model generates predictions through a series of decision rules based on health attributes. Random Forest combines multiple decision trees to improve prediction accuracy and reduce the possibility of overfitting. The use of multiple algorithms allows the system to compare different predictive approaches and identify the most suitable model.

Following model development, each machine learning model is evaluated and compared using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Accuracy measures the overall correctness of the predictions, precision determines the proportion of correctly predicted high-risk cases, recall evaluates the model's ability to identify actual high-risk patients, and F1-score provides a balanced measure of precision and recall.

The confusion matrix offers detailed insights into correctly and incorrectly classified cases. Based on these evaluation metrics, the best-performing model is selected for deployment.

To enhance transparency and interpretability, SHAP (SHapley Additive Explanations) is applied to the selected model. SHAP explains how individual features contribute to the prediction outcome by assigning importance values to each attribute. This enables users and healthcare professionals to understand the factors that increase or decrease the predicted heart attack risk. For example, factors such as high cholesterol levels, elevated blood pressure, may contribute positively to risk prediction, while healthy lifestyle factors may reduce the predicted risk.

Once the best-performing model is integrated with SHAP analysis, the system is capable of processing new patient data. When a patient's health information is entered into the system, the trained model analyzes the input and predicts whether the patient falls into a low-risk or high-risk category. Along with the prediction result, SHAP generates a detailed explanation showing the contribution of each feature toward the final decision. This combination of prediction and explanation makes the system more transparent and trustworthy.

The system also includes a storage module that maintains records of patient information, prediction outcomes and prediction timestamps. Storing this information enables future analysis, patient monitoring, and helps maintain accurate patient histories.

Overall, the proposed Heart Attack Risk Prediction System integrates machine learning and explainable artificial intelligence to provide accurate and interpretable predictions.

3.2 Workflow of the Project.

The workflow of the proposed Heart Attack Risk Prediction System begins with collecting the Heart Disease Dataset from the UCI Machine Learning Repository and performing data preprocessing to clean and prepare the data for analysis. The processed dataset is then divided into training and testing sets. Multiple machine learning algorithms, including Logistic Regression, SVM, KNN, Decision Tree, and Random Forest, are trained and evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix. The best-performing model is selected and integrated with SHAP to explain the contribution of each feature to the prediction. When new patient data is entered, the system predicts the heart attack risk and generates a feature-wise explanation. Finally, the prediction results and patient information are stored for future reference and monitoring.

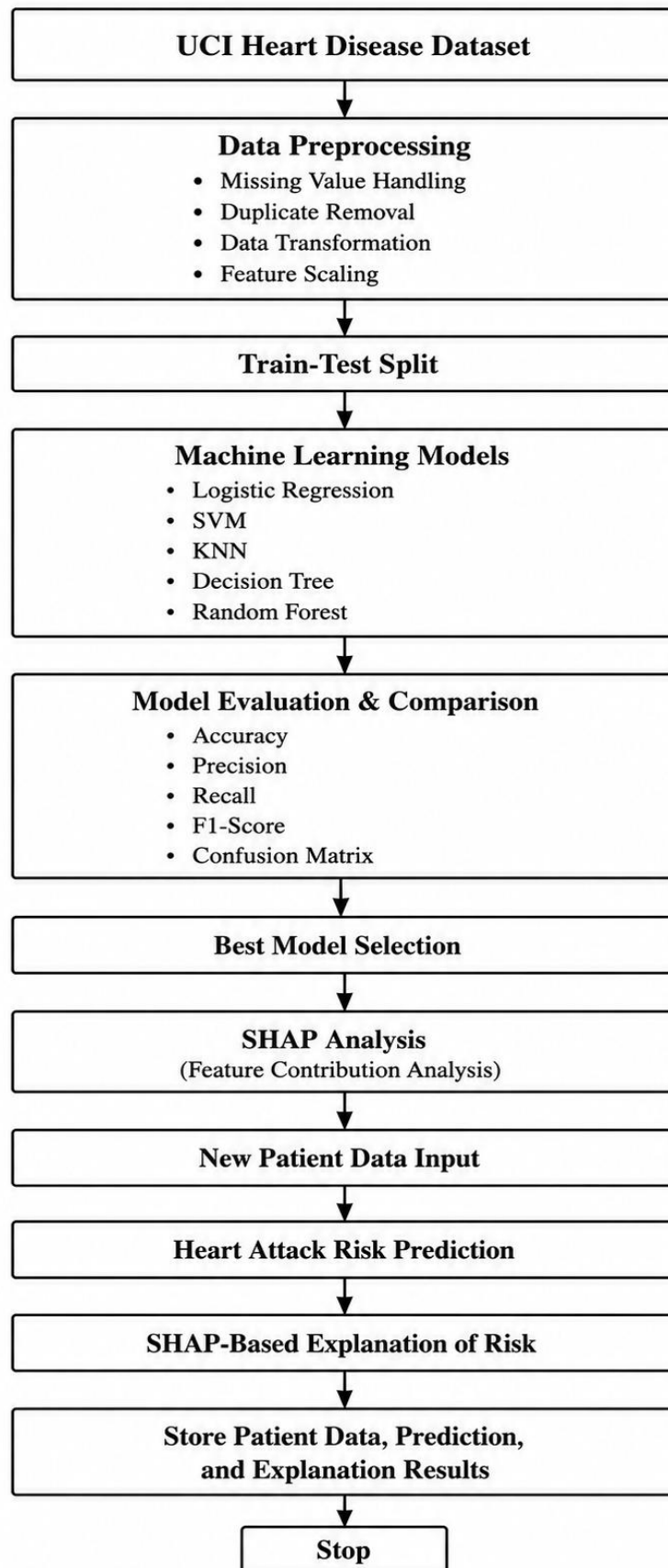


Figure 3.2: Workflow for Heart Attack Prediction

3.2.1 Dataset collection

The workflow begins with the collection of the **Heart Disease Dataset from the UCI Machine Learning Repository**, which is a widely used dataset for cardiovascular disease prediction research. The dataset contains patient health information and clinical attributes that are associated with heart disease and heart attack risk. The dataset contains **303 patient records** and **14 attributes**, including **13 input features and 1 target variable**. It consists of mixed data types, including numerical, categorical, and binary attributes.. The dataset serves as the primary source of information for training and testing the machine learning models.

The attributes provide valuable information that helps machine learning algorithms identify patterns associated with heart disease risk.

- **Data description .**

The dataset contains patient health information and clinical attributes that are associated with heart disease and heart attack risk. The dataset contains **303 patient records** and **14 attributes**, including **13 input features and 1 target variable**.

Feature Name	Description
age	Age of the patient in years
sex	Gender of the patient (1 = male, 0 = female)
cp	Chest pain type (0 = typical angina, 1 = atypical angina, 2 = non-anginal pain, 3 = asymptomatic)
trtbps	Resting blood pressure (in mm Hg)
chol	Serum cholesterol level (mg/dl)
fbs	Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)
restecg	Resting electrocardiographic results (0 = normal, 1 = ST-T abnormality, 2 = LV hypertrophy)
thalachh	Maximum heart rate achieved
exng	Exercise induced angina (1 = yes, 0 = no)
oldpeak	ST depression induced by exercise relative to rest
slp	Slope of peak exercise ST segment (0 = downsloping, 1 = flat, 2 = upsloping)
caa	Number of major vessels colored by fluoroscopy (0–3)

thall	Thalassemia test result (1 = fixed defect, 2 = normal, 3 = reversible defect)
output	Target variable (0 = low/no heart disease risk, 1 = high heart disease risk)

Table 3.2.1 Dataset description

3.2.2. Data Preprocessing

Data preprocessing is one of the most important stages in the proposed Heart Attack Risk Prediction System, as the quality of the input data directly affects the performance and accuracy of machine learning models. Real-world healthcare datasets often contain missing values, duplicate records, inconsistent entries, and data quality issues that can negatively impact the prediction process. Therefore, preprocessing is performed to transform the raw dataset into a clean, structured, and machine-learning-ready format.

- **Handling Missing Values**

Missing values occur when information for one or more attributes is unavailable in certain records. Such incomplete data can affect model training and lead to inaccurate predictions. Therefore, the dataset is carefully examined to identify missing values.

- **Removing Duplicate Records**

Duplicate records refer to repeated patient entries within the dataset. These duplicates can introduce bias and cause machine learning models to learn incorrect patterns.

- **Correcting Inconsistent Data Entries**

Healthcare datasets may contain inconsistencies due to manual data entry errors, formatting differences, or incorrect values. For example, variations in categorical values, invalid numerical entries, or mismatched formats can create confusion for machine learning algorithms.

- **Feature Scaling and Normalization:**

feature scaling techniques such as normalization or standardization are applied where necessary. Scaling ensures that all features contribute proportionally to the learning process and helps improve the performance of algorithms such as SVM and KNN.

3.2.3. Train-Test Split

After data preprocessing, the dataset is divided into two subsets known as the **training set** and the **testing set**. This process is called the **train-test split** and is an important step in machine learning model development. The main purpose of splitting the dataset is to evaluate how well a model performs on unseen data and to ensure that it can generalize effectively to new patient records.

- **Training set:** is used to train the machine learning algorithms by allowing them to learn patterns and relationships between the input features and the target variable. In the proposed system, **80% of the dataset** is allocated for training. During this phase, the model analyzes patient attributes such as age, blood pressure, cholesterol level, and other health indicators to understand how these factors influence heart attack risk and to develop predictive patterns.
- **Testing set:** is used after training to evaluate the model's performance. The remaining **20% of the dataset** is reserved for testing and is not exposed to the model during the training process. Since the testing data is not used during the training process, it provides an unbiased assessment of how accurately the model can predict heart attack risk for new patients. This helps determine whether the model has learned meaningful patterns or simply memorized the training data

3.2.4 machine learning models

machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed. Instead of following fixed rules, machine learning algorithms analyze historical data, identify relationships among variables, and use the learned patterns to predict outcomes for new data.

Machine learning can be broadly classified into three main types:

- **Supervised Learning**
Supervised learning uses labeled data, where both input features and the corresponding output values are available.

Examples: Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Random Forest.

- **Unsupervised Learning**

Unsupervised learning works with unlabeled data, where the output variable is unknown. The objective is to discover hidden patterns, relationships, or groupings within the data.

Examples: K-Means Clustering, Hierarchical Clustering, Principal Component Analysis (PCA).

- **Reinforcement Learning**

Reinforcement learning enables an agent to learn through interaction with an environment. The agent receives rewards or penalties based on its actions and gradually learns the best strategy to achieve a goal.

Examples: Robotics, Game Playing Systems, Autonomous Vehicles.

- **Logistic Regression**

Logistic Regression is a supervised classification algorithm used to predict the probability of a patient being at risk of a heart attack. It analyzes the relationship between patient attributes and the target variable and produces a binary output indicating low or high risk. Due to its simplicity and effectiveness, it is often used as a baseline model.

- **Support Vector Machine (SVM)**

Support Vector Machine is a supervised classification algorithm that separates data into different classes by finding the optimal boundary, known as a hyperplane. It is effective for handling complex datasets and can achieve high prediction accuracy by maximizing the distance between classes.

- **K-Nearest Neighbors (KNN)**

K-Nearest Neighbors is a supervised learning algorithm that classifies a patient based on similarities with other patients in the dataset. It identifies the nearest neighboring

records and assigns the most common class among them. The algorithm is simple and useful for pattern recognition.

- **Decision Tree**

Decision Tree is a supervised classification algorithm that uses a tree-like structure to make decisions. Each node represents a condition based on a feature, and the branches represent possible outcomes. The final prediction is obtained by traversing the tree from the root node to a leaf node. It is highly interpretable and easy to visualize.

3.2.5 Model Evaluation, Comparison, and Best Model Selection

After training the machine learning models, their performance is evaluated and compared to identify the most suitable algorithm for heart attack risk prediction. The models used in this study include Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, and Random Forest. Each model is assessed using performance metrics such as **Accuracy, Precision, Recall, F1-Score, and Confusion Matrix**. Confusion Matrix Elements. The confusion matrix has four elements:

- (1) True positives (TP): the number of instances where the model correctly predicts the positive outcome
 - (2) False positives (FP): the number of instances where the model incorrectly predicts the positive outcome
 - (3) True negatives (TN): the number of instances where the model correctly predicts the negative outcome
 - (4) False negatives (FN): the number of instances where the model incorrectly predicts the negative outcome
- 1. Accuracy**

Accuracy measures how many total predictions the model got correct out of all predictions.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

2. Precision

Precision measures how many of the predicted positive cases are actually correct.

Formula:

$$Precision = \frac{TP}{TP + FP}$$

Recall (Sensitivity)

Recall measures how many actual positive cases were correctly identified by the model.

Formula:

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

F1-score is the balance between precision and recall. It is used when both false positives and false negatives are important.

Formula:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The performance results of all models are compared based on these evaluation metrics. The algorithm that achieves the highest overall performance and provides the most reliable predictions is selected as the final model. The selected best-performing model is then used for heart attack risk prediction and integrated with SHAP analysis to explain the contribution of individual features to the prediction outcome. This process ensures that the system utilizes the most accurate and dependable model for predicting heart attack risk.

3.2.6 SHAP Analysis (Feature Contribution Analysis)

After selecting the best-performing machine learning model, **SHAP (SHapley Additive Explanations)** is applied to improve the interpretability and transparency of the prediction process. SHAP is an Explainable Artificial Intelligence (XAI) technique that explains how each feature contributes to the model's prediction. Instead of providing only the prediction result, SHAP helps identify the factors that influence the prediction and their level of impact.

3.2.7 New Patient Data Input, Risk Prediction, and SHAP Analysis

After selecting the best-performing machine learning model, the system is ready to process new patient information. In this stage, the patient enters health-related details such as age, sex, chest pain type, blood pressure, cholesterol level, fasting blood sugar, maximum heart rate, exercise-induced angina, and other clinical attributes through the user interface. The entered data undergoes the same preprocessing steps used during model training to ensure consistency and accuracy.

Once the data is processed, it is provided to the selected machine learning model for prediction. The model analyzes the patient's health attributes and predicts whether the individual has a **low risk** or **high risk** of heart attack. This prediction is generated based on the patterns learned from the training dataset and helps in the early identification of patients who may require medical attention.

After the prediction is made, **SHAP (SHapley Additive Explanations)** is applied to explain the result. SHAP calculates the contribution of each feature toward the prediction and identifies the factors that increase or decrease the patient's heart attack risk. The explanation highlights the most influential attributes and their impact on the final outcome. By combining risk prediction with SHAP analysis, the system provides both accurate predictions and clear explanations, enabling healthcare professionals and patients to better understand the reasons behind the predicted risk level.

3.2.8 Store Patient Data and Prediction

After the heart attack risk prediction and SHAP analysis are completed, the system stores the patient's information and prediction results for future reference and analysis. The stored data includes patient health attributes, the predicted risk level, prediction probability, and the corresponding SHAP explanation values. Maintaining these records helps in tracking patient history, monitoring changes in health conditions, and supporting future medical assessments.

3.3 Algorithm design :

An algorithm is a well-defined sequence of step-by-step instructions used to solve a problem or accomplish a specific task. In computer science and machine learning, algorithms provide a systematic approach for processing data, performing calculations, and generating meaningful results. They help ensure that tasks are executed efficiently, accurately, and consistently. In the proposed Heart Attack Risk Prediction System, algorithms are used to analyze patient health data, identify patterns related to heart disease risk, and generate predictions based on learned relationships within the dataset. Machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), AdaBoost, Gradient Boosting, and XGBoost are employed to classify patients as either high-risk or low-risk. Additionally, algorithms are used for data preprocessing, model optimization through GridSearchCV, performance evaluation, and SHAP-based explanation generation.

Thus, algorithms form the core of the system by enabling accurate prediction, efficient data processing, and interpretable decision-making in healthcare applications.

Steps:

1. **Load the UCI Heart Disease Dataset**

- Import the dataset containing patient demographic, clinical, and diagnostic attributes.
- Identify the target variable indicating heart disease risk.

2. **Perform Data Preprocessing**

- Handle missing values and duplicate records.
- Encode categorical features into numerical form.
- Normalize or scale numerical attributes where required.
- Prepare the dataset for machine learning analysis.

3. **Perform Exploratory Data Analysis (EDA)**

- Analyze feature distributions and correlations.
- Identify important risk factors associated with heart disease.
- Visualize relationships between features and the target variable.

4. **Split the Dataset**

- Divide the dataset into training and testing sets.
- Use the training set for model learning and the testing set for evaluation.

5. **Define the Machine Learning Models**

- Logistic Regression (LR)
- Decision Tree (DT)
- Random Forest (RF)
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- AdaBoost (AB)
- Gradient Boosting (GB)
- Extreme Gradient Boosting (XGBoost)

6. **For Each Model**

- Train the model using the training dataset.
- Apply **GridSearchCV** for hyperparameter tuning and cross-validation.
- Optimize model parameters to improve predictive performance.
- Evaluate the trained model using the testing dataset.

- Record evaluation metrics such as:
 - Accuracy
 - Precision
 - Recall
 - F1-Score

7. Compare Model Performance

- Compare the results of all trained models based on the evaluation metrics.
- Analyze strengths and weaknesses of each model.

8. Select the Best-Performing Model

- Identify the model with the highest overall performance.
- Save the selected model for future predictions.

9. Perform SHAP Analysis

- Apply SHAP (SHapley Additive Explanations) to the selected model.
- Determine the contribution of each feature to the prediction.
- Generate both global and local feature importance explanations.

10. Predict Heart Attack Risk for New Patients

- Accept new patient health information as input.
- Process the input using the trained model.
- Predict whether the patient is at low risk or high risk of heart disease.
- Display the prediction result along with the probability score.

11. Generate Feature Contribution Explanation

- Use SHAP to explain how each feature influenced the prediction.
- Highlight the factors that increased or decreased the predicted risk.

12. Store Prediction Results

- Save patient details, prediction outcomes, probability scores, and SHAP explanations in a CSV file or database.
- Maintain records for future analysis and monitoring.

End

3.4 Summarization Models

3.4.1 Logistic Regression (LR)

Logistic Regression is one of the most widely used supervised machine learning algorithms for binary classification problems, especially in healthcare applications. In a heart attack prediction system, Logistic Regression analyzes the relationship between patient health attributes and the likelihood of developing a heart attack. The algorithm assigns different levels of importance to each feature, such as age, cholesterol level, blood pressure, diabetes status, smoking habits, and body mass index. Based on these relationships, it estimates the probability that a patient belongs to either the high-risk or low-risk category. Logistic Regression is highly valued because its predictions are easy to interpret, allowing healthcare professionals to understand how each factor influences the outcome. Although it performs well when the relationship between features and the target variable is relatively linear, its predictive capability may decrease when dealing with highly complex and non-linear medical data.

3.4.2 Decision Tree (DT)

A Decision Tree is a classification model that mimics human decision-making by breaking down a complex problem into a series of simple decisions. The model begins with the entire dataset and repeatedly divides it into smaller groups based on the most informative patient attributes. Each branch of the tree represents a decision rule, while the leaf nodes represent the final prediction. In heart attack prediction, the model may first split patients based on age, then cholesterol level, blood pressure, or other risk factors until a final classification is reached. One of the major advantages of Decision Trees is their interpretability, as the reasoning process behind each prediction can be easily visualized and understood. This transparency makes them particularly useful in healthcare environments where explanations are important. However, Decision Trees can become overly complex and may memorize training data, resulting in reduced performance on unseen patient records.

3.4.3 Random Forest (RF)

Random Forest is an ensemble learning technique that improves upon the limitations of a single Decision Tree by combining the predictions of multiple trees. Instead of relying on one tree, the algorithm generates numerous Decision Trees using different subsets of training data and features. Each tree independently predicts whether a patient is at risk of a heart attack, and the final prediction is determined through majority voting among all trees. This collective

decision-making process significantly improves accuracy, stability, and robustness. Random Forest is capable of handling large datasets with many features and can effectively capture complex interactions among health indicators. It also reduces the risk of overfitting that commonly affects individual Decision Trees. Due to its strong predictive performance and reliability, Random Forest is frequently used in medical diagnosis and disease risk prediction systems.

3.4.4 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a simple yet effective machine learning algorithm that classifies a patient based on the characteristics of similar patients in the dataset. Rather than building a mathematical model during training, KNN stores all available training examples and uses them during prediction. When a new patient record is provided, the algorithm identifies the most similar patients based on their health attributes. The patient's risk category is then determined according to the majority class among these neighboring cases. In heart attack prediction, patients with similar age, cholesterol levels, blood pressure readings, and lifestyle habits are grouped together to estimate risk. KNN is easy to implement and can produce accurate results when the dataset is well-structured. However, its performance can decline when dealing with large datasets because every prediction requires comparison with many existing records.

3.4.5 Support Vector Machine (SVM)

Support Vector Machine is a powerful supervised learning algorithm designed to find the optimal boundary that separates different classes within a dataset. In heart attack prediction, SVM analyzes patient records and determines the best possible division between high-risk and low-risk individuals. The algorithm focuses on the most critical data points, known as support vectors, which lie closest to the decision boundary. By maximizing the separation between classes, SVM aims to improve generalization and reduce classification errors. One of its major strengths is its ability to handle complex datasets where relationships between features are not straightforward. Through the use of kernel functions, SVM can transform data into higher-dimensional spaces and identify non-linear patterns that may not be visible in the original feature space. This capability makes SVM particularly suitable for medical datasets containing intricate relationships among risk factors.

3.4.6 AdaBoost (Adaptive Boosting)

AdaBoost is a boosting-based ensemble learning algorithm that combines multiple weak classifiers to create a highly accurate predictive model. The algorithm operates by training a sequence of simple models, usually shallow Decision Trees, where each new model focuses on correcting the mistakes made by previous models. During training, greater importance is assigned to patient records that were incorrectly classified, forcing subsequent learners to pay more attention to difficult cases. As the learning process continues, the algorithm gradually improves its ability to distinguish between high-risk and low-risk patients. The final prediction is generated by combining the weighted contributions of all individual classifiers. AdaBoost often achieves higher accuracy than a single Decision Tree because it systematically reduces classification errors. However, it may be sensitive to noisy or inconsistent medical records.

3.4.7 Gradient Boosting (GB)

Gradient Boosting is an advanced ensemble learning technique that builds a strong predictive model through a sequence of weak learners. Unlike AdaBoost, which focuses on misclassified samples, Gradient Boosting concentrates on minimizing the prediction errors generated by previous models. At each stage, a new Decision Tree is trained specifically to correct the residual errors remaining from earlier predictions. This iterative process continues until the model achieves optimal performance or reaches a predefined stopping criterion. In heart attack prediction, Gradient Boosting effectively captures complex relationships among various health indicators, enabling highly accurate risk assessment. The algorithm is capable of modeling subtle interactions between patient characteristics and disease outcomes, making it one of the most effective techniques for structured healthcare datasets. However, proper parameter tuning is necessary to avoid overfitting and excessive training times.

3.4.8 XGBoost (Extreme Gradient Boosting)

XGBoost is an enhanced implementation of Gradient Boosting that incorporates several optimization techniques to improve efficiency, speed, and predictive performance. It has become one of the most popular machine learning algorithms for classification and prediction tasks due to its exceptional accuracy. XGBoost builds decision trees sequentially while applying regularization techniques that help prevent overfitting. The algorithm includes advanced features such as parallel processing, automatic handling of missing values, and efficient memory utilization, making it suitable for large-scale medical datasets. In heart attack

prediction systems, XGBoost can identify complex patterns among risk factors and generate highly reliable predictions. Its ability to balance model complexity with generalization often results in superior performance compared to traditional machine learning algorithms. Because of these advantages, XGBoost is frequently selected as the best-performing model during comparative evaluations of heart disease prediction systems.

3.5 Implementation

The implementation of the proposed Heart Attack Risk Prediction System is carried out through a sequence of stages that transform raw patient health data into accurate and interpretable risk predictions. The system combines machine learning algorithms with Explainable Artificial Intelligence (XAI) techniques to not only predict heart attack risk but also explain the factors influencing each prediction. The implementation begins with data acquisition and preprocessing, followed by exploratory analysis, model training, evaluation, prediction generation, SHAP-based interpretation, and storage of patient records.

3.5.1 Data collecting and Preprocessing

The implementation starts with the UCI Heart Disease dataset, which contains patient clinical, and diagnostic information. Since machine learning models require clean and consistent data, preprocessing is performed to remove duplicate records, handle missing values, and correct inconsistencies. Categorical attributes such as sex, chest pain type, exercise-induced angina, and thalassemia results are encoded into numerical form so that they can be processed by machine learning algorithms. Numerical features such as age, cholesterol level, blood pressure, and maximum heart rate are analyzed and scaled when necessary. This preprocessing stage improves data quality and ensures that the models learn meaningful patterns from the dataset.

3.5.2 Exploratory Data Analysis (EDA)

After preprocessing, Exploratory Data Analysis is conducted to understand the characteristics of the dataset. Statistical summaries and visualizations are generated to examine feature distributions, identify correlations among variables, and detect outliers. Correlation analysis helps determine which factors have the strongest relationship with heart disease risk. For example, cholesterol levels, blood pressure, chest pain type, and exercise-induced angina may show significant associations with the target variable. This stage provides valuable insights into the dataset and supports informed model development.

3.5.3 Dataset Splitting

The processed dataset is divided into training and testing datasets. Typically, a larger portion of the data is allocated for training while the remaining portion is reserved for testing. The training dataset is used to teach the machine learning models, whereas the testing dataset is used to evaluate their performance on unseen data. This separation helps measure how well the models generalize to new patient records.

3.5.4 Model Training

Multiple machine learning classification algorithms are trained using the prepared dataset. These include Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), AdaBoost, Gradient Boosting, and XGBoost. Each algorithm learns the relationship between patient health attributes and the likelihood of heart disease. During training, the models identify patterns within the data that distinguish high-risk patients from low-risk patients. Since different algorithms have different learning mechanisms, training multiple models allows a comprehensive comparison of their predictive capabilities.

3.5.5 Hyperparameter Optimization Using GridSearchCV

To improve model performance, GridSearchCV is applied during training. GridSearchCV systematically evaluates different combinations of hyperparameters for each algorithm using cross-validation. For example, it can test different numbers of trees in Random Forest, different kernel functions in SVM, or various learning rates in boosting algorithms. By selecting the best-performing parameter combination, GridSearchCV helps maximize prediction accuracy and improve model generalization.

3.5.6 Model Evaluation and Comparison

After training, all models are evaluated using the testing dataset. Performance is measured using evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC Score. Accuracy measures the overall correctness of predictions, Precision evaluates the correctness of positive predictions, Recall measures the ability to identify actual high-risk patients, and F1-Score provides a balance between Precision and Recall. The performance results of all models are compared, and the model with the highest overall performance is selected as the final prediction model.

3.5.7 Heart Attack Risk Prediction

Once the best-performing model is selected, it is integrated into the prediction system. Users can enter new patient information through the system interface, including age, sex, blood pressure, cholesterol level, fasting blood sugar, chest pain type, and other health-related attributes. The trained model processes these inputs and predicts whether the patient has a low or high risk of heart disease. Along with the classification result, the model can also provide a probability score indicating the confidence level of the prediction.

3.5.8 SHAP-Based Explainability

To enhance transparency and trust, SHAP (SHapley Additive Explanations) is incorporated into the system. After generating a prediction, SHAP calculates the contribution of each feature toward the final result. For example, a high cholesterol level, advanced age, or exercise-induced angina may increase the predicted risk, while favorable health indicators may reduce it. SHAP produces both global explanations, showing overall feature importance across the dataset, and local explanations, explaining the prediction for an individual patient. This allows healthcare professionals to understand why a specific prediction was made rather than relying on a black-box result.

3.5.9 Storage of Patient Records and Predictions

The system includes a storage module that saves newly entered patient information along with prediction outcomes and SHAP explanations. The records are stored in a separate CSV file or database for future reference. Each stored entry typically contains patient details, prediction results, probability scores, date and time of prediction, and feature contribution values. Maintaining these records supports future analysis, monitoring, reporting, and potential model retraining.

3.5.10 System Outcome

The final implemented system provides an intelligent and explainable solution for heart attack risk prediction. By integrating multiple machine learning algorithms, hyperparameter optimization, comprehensive evaluation, SHAP-based interpretation, and automated record storage, the system offers accurate predictions and meaningful explanations. This helps healthcare professionals identify high-risk patients at an early stage, supports clinical decision-

making, and promotes preventive healthcare strategies aimed at reducing cardiovascular disease-related complications.

3.6 Summary

This chapter presented the **Proposed Methodology and Implementation** of the Heart Attack Risk Prediction System. Section 3.1 discussed the proposed methodology adopted for predicting heart attack risk using machine learning techniques. Section 3.2 explained the workflow of the project, including data preprocessing, exploratory data analysis, model training, evaluation, prediction, and SHAP-based explanation generation. Section 3.3 described the UCI Cleveland Heart Disease dataset, including its features, attribute types, and target variable used for classification. Section 3.4 provided a summary of the machine learning models used in the system, namely Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), AdaBoost, Gradient Boosting, and XGBoost. Finally, Section 3.5 explained the implementation process, including hyperparameter tuning using GridSearchCV, model evaluation, best model selection, prediction of new patient data, SHAP analysis, and storage of prediction results. The concepts discussed in this chapter establish the foundation for the development and performance evaluation of the proposed heart attack risk prediction system.

4. RESULT AND ANALYSIS

This chapter presents the results and analysis of the proposed Heart Attack Risk Prediction System. Multiple machine learning models, including Logistic Regression, Decision Tree, Random Forest, KNN, SVM, AdaBoost, Gradient Boosting, and XGBoost, were trained and evaluated using the UCI Cleveland Heart Disease dataset. Their performance was compared using metrics such as Accuracy, Precision, Recall, F1-Score to identify the best-performing model. GridSearchCV was used to optimize model performance through hyperparameter tuning. Additionally, SHAP analysis was applied to explain the contribution of individual features to the prediction results. The findings demonstrate that the proposed system provides accurate and interpretable heart disease risk predictions, supporting effective clinical decision-making.

1. Experimental Setup

The experiments were conducted using the preprocessed UCI Cleveland Heart Disease dataset. The dataset was divided into training and testing subsets to evaluate the ability of the models to generalize to unseen patient data. Multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), AdaBoost, Gradient Boosting, and XGBoost, were implemented and trained. Hyperparameter tuning was performed using GridSearchCV to obtain the optimal model configuration and improve predictive performance.

2 Model Performance Analysis

The performance of each machine learning model was evaluated and compared to determine its effectiveness in classifying patients into high-risk and low-risk categories. The results showed that all models were capable of identifying patterns related to heart disease from patient health records. However, differences in predictive performance were observed among the models.

Traditional machine learning algorithms such as Logistic Regression and Decision Tree provided satisfactory results and were relatively easy to interpret. KNN and SVM also demonstrated good classification performance but required careful parameter tuning for optimal results. Ensemble learning techniques, including Random Forest, AdaBoost, Gradient Boosting, and XGBoost, generally outperformed individual classifiers due to their ability to combine multiple learning models and capture complex relationships among features. These

ensemble methods produced more stable and accurate predictions, making them highly suitable for healthcare applications.

After comparing the results of all models, the best-performing model was selected based on its overall performance across multiple evaluation metrics. This model was subsequently used as the final prediction model in the proposed system.

3 Evaluation Metrics Analysis

Several evaluation metrics were used to assess model performance comprehensively. Accuracy measured the proportion of correctly classified patient records among all predictions. Precision evaluated the reliability of positive predictions by determining how many patients predicted to be at risk were actually at risk. Recall measured the model's ability to correctly identify patients who truly had heart disease risk, which is especially important in healthcare applications where missed diagnoses can have serious consequences.

The F1-Score provided a balanced measure of Precision and Recall, making it useful when both false positives and false negatives need to be considered. The evaluation results indicated that the selected model maintained strong and balanced performance across all metrics, demonstrating excellent predictive capability and reliability.

4 SHAP Analysis

To enhance transparency and interpretability, SHAP analysis was integrated into the proposed system. SHAP provided insights into how individual features contributed to the model's predictions. Both global and local explanations were generated to improve understanding of the model's behavior.

The global SHAP analysis identified the most influential features affecting heart disease prediction across the entire dataset. Features such as chest pain type, age, cholesterol level, maximum heart rate achieved, exercise-induced angina, number of major vessels, and ST depression were found to have a significant impact on prediction outcomes. Local SHAP explanations were generated for individual patients, illustrating how specific feature values increased or decreased the predicted heart disease risk. These explanations improved trust in the model by providing clear reasoning behind each prediction.

5 New Patient Prediction Analysis

The developed system was tested using newly entered patient records through the user interface. After receiving patient information, the selected machine learning model generated a heart disease risk prediction along with a probability score indicating the likelihood of risk. Simultaneously, SHAP explanations were produced to highlight the contribution of each feature toward the prediction result.

6 Overall Analysis

The results obtained from the experiments demonstrate that the proposed Heart Attack Risk Prediction System effectively predicts heart disease risk using machine learning techniques. The comparison of multiple classification algorithms enabled the identification of the most suitable model, while GridSearchCV improved performance through hyperparameter optimization. The integration of SHAP enhanced the interpretability and transparency of predictions by explaining feature contributions. Overall, the system achieved reliable predictive performance, provided meaningful explanations, and showed significant potential as a clinical decision-support tool for early heart disease detection and preventive healthcare management.

4.1 Experimental Results.

4.1.1 Dataset Analysis

The experiments were conducted using the **UCI Heart Disease dataset**, which contains 14 attributes, including 13 input features and one target variable for heart disease prediction. Before training, the dataset was preprocessed by handling missing values, removing duplicates, and encoding categorical features to improve data quality.

	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1

Figure 4.1.1 Dataset analysis

The dataset was divided into **training and testing sets**, with 80% of the data used for training and 20% for testing..

Train: 0.8502202643171806

Test : 0.8947368421052632

The implementation was carried out using **Python** and libraries such as **Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, and SHAP** for data processing, model development, evaluation, visualization, and explainability.

4.1.2 Correlation Analysis

Correlation analysis was performed to examine the relationships between the features in the UCI Cleveland Heart Disease dataset and the target variable. A **correlation matrix** was generated to calculate the correlation coefficient between each pair of features, helping to identify how strongly they are related to one another. The correlation values range from **-1 to +1**, where positive values indicate a direct relationship and negative values indicate an inverse relationship.

To provide a clear visual representation, a **heatmap** was created using the correlation matrix. The heatmap highlights the strength and direction of correlations through varying color intensities, making it easier to identify important relationships among features. Features with stronger correlations appear more prominently, allowing for better understanding of the dataset structure.

The analysis revealed that several attributes showed a significant relationship with the target variable (heart disease risk). Features such as **chest pain type (cp)**, **exercise-induced angina (exng)**, **number of major vessels (caa)**, **thalassemia (thall)**, and **oldpeak** exhibited positive correlations with heart disease, indicating that higher values of these attributes are associated with an increased risk of heart disease. On the other hand, features such as **maximum heart rate achieved (thalachh)** generally showed a negative correlation with the target variable, suggesting that higher heart rates may be associated with a lower likelihood of heart disease in the dataset.

Overall, correlation analysis provided valuable insights into the relationships between patient attributes and heart disease risk. It helped identify influential features, supported feature understanding, and contributed to the development of more effective machine learning models for heart disease prediction.

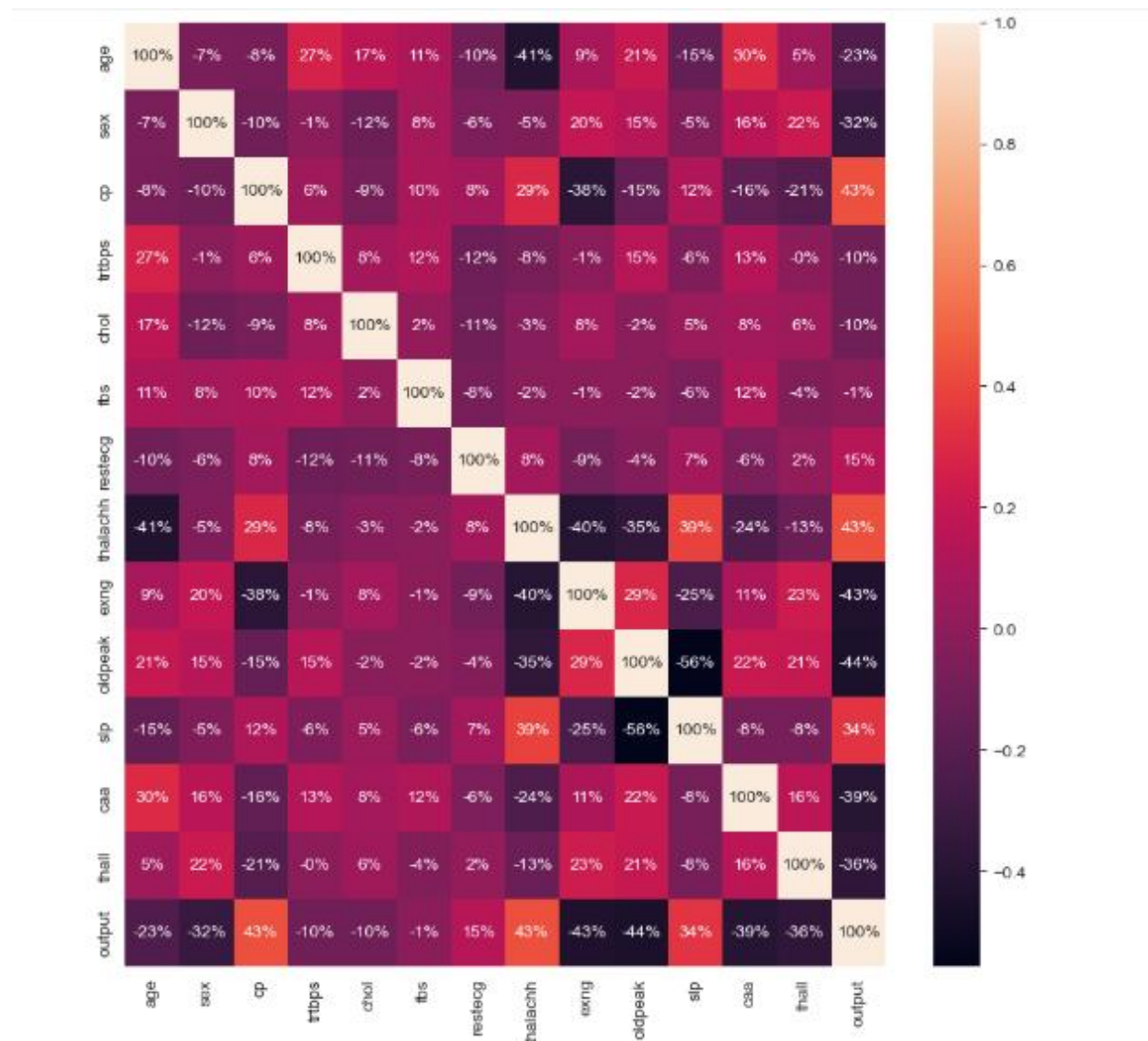


Figure 4.1.2 Correlation Analysis

4.1.3 Exploratory Data Analysis (EDA)

Age Distribution Analysis

The age distribution analysis was performed to understand the demographic characteristics of patients included in the dataset. The distribution plot shows that patient ages are spread across different age groups, with a higher concentration of individuals in the middle-aged and older adult categories. Since cardiovascular diseases are more common among older individuals, the presence of a larger number of patients in these age groups is expected. The analysis helps identify whether the dataset contains a balanced representation of different age ranges and provides insights into the relationship between age and heart attack risk.

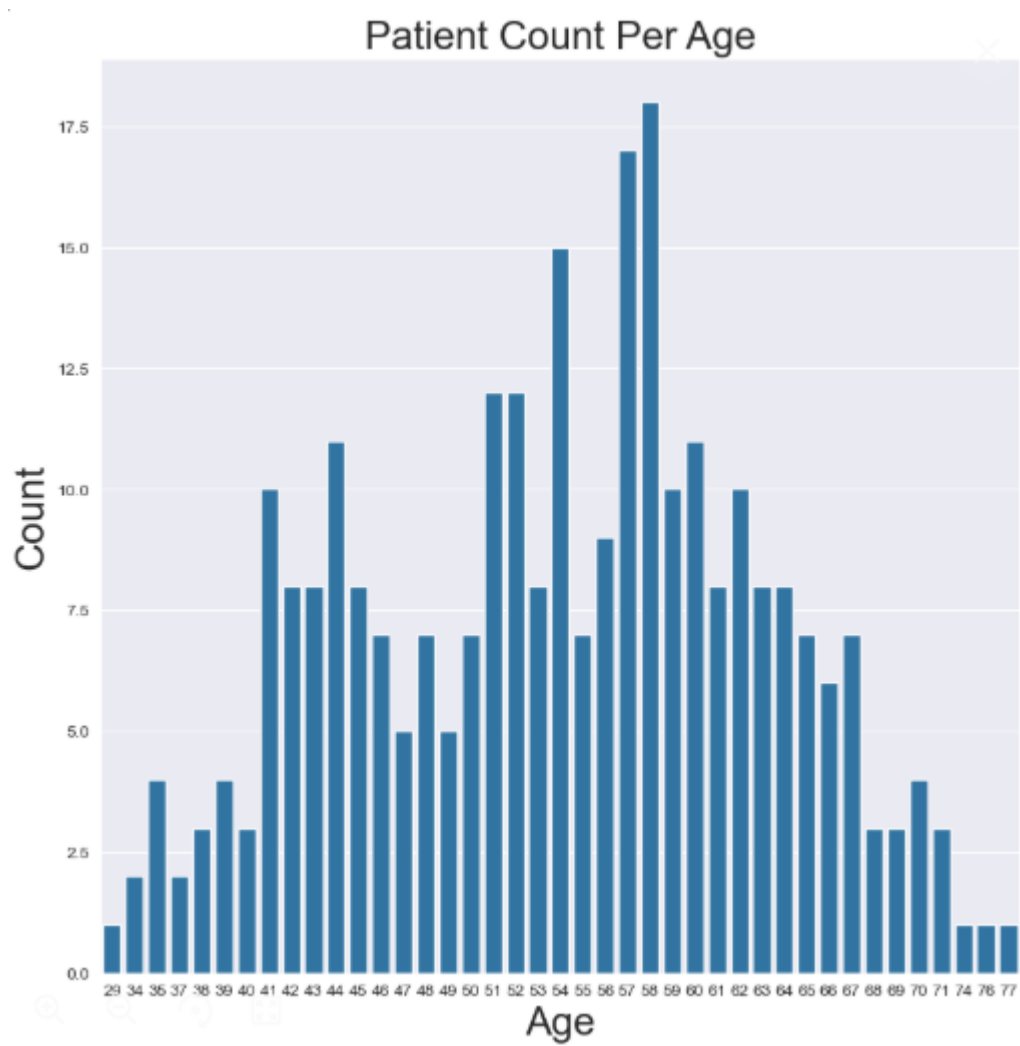


Figure 4.1.3 Age Distribution Analysis

4.1.4 Classification Report

The performance of the **Logistic Regression** model was evaluated using standard classification metrics, including Precision, Recall, F1-Score, and Accuracy. The model achieved an overall **accuracy of 89%**, indicating its ability to correctly classify the majority of patient records. For the low-risk class, the model obtained a precision of **92%**, recall of **85%**, and F1-score of **88%**, demonstrating strong performance in identifying patients with lower heart disease risk. For the high-risk class, the model achieved a precision of **88%**, recall of **93%**, and F1-score of **90%**. The high recall value is particularly significant in healthcare applications, as it shows that the model successfully identified most patients who were actually at risk of heart disease, thereby reducing the likelihood of missed diagnoses. Overall, the results indicate that the Logistic Regression model provides reliable and balanced performance, making it a suitable approach for heart disease risk prediction.

	precision	recall	f1-score	support
Collapse Output	0.92	0.85	0.88	27
1	0.88	0.93	0.90	30
accuracy			0.89	57
macro avg	0.90	0.89	0.89	57
weighted avg	0.90	0.89	0.89	57

Figure 4.1.4 Classification Report

4.1.5 Confusion Matrix Analysis

The confusion matrix obtained from the Logistic Regression model is shown in Figure 4.X. The matrix indicates that **23 low-risk patients (Class 0)** were correctly classified as low risk (**True Negatives**), while **28 high-risk patients (Class 1)** were correctly classified as high risk (**True Positives**). Additionally, **4 low-risk patients** were incorrectly predicted as high risk (**False Positives**), and **2 high-risk patients** were incorrectly classified as low risk (**False Negatives**).

The results demonstrate that the model performs well in distinguishing between high-risk and low-risk patients. The low number of **False Negatives (2)** is particularly important in healthcare applications, as it indicates that only a small number of patients with actual heart disease risk were missed by the model. Similarly, the number of **False Positives (4)** is relatively low, showing that the model does not frequently classify healthy patients as high risk.

Overall, the confusion matrix confirms the effectiveness of the Logistic Regression model, achieving **89% classification accuracy** with a strong ability to identify patients at risk of heart disease. The model correctly classified **51 out of 57 patient records**, demonstrating reliable performance for heart disease risk prediction and supporting its use as a decision-support tool in healthcare applications.

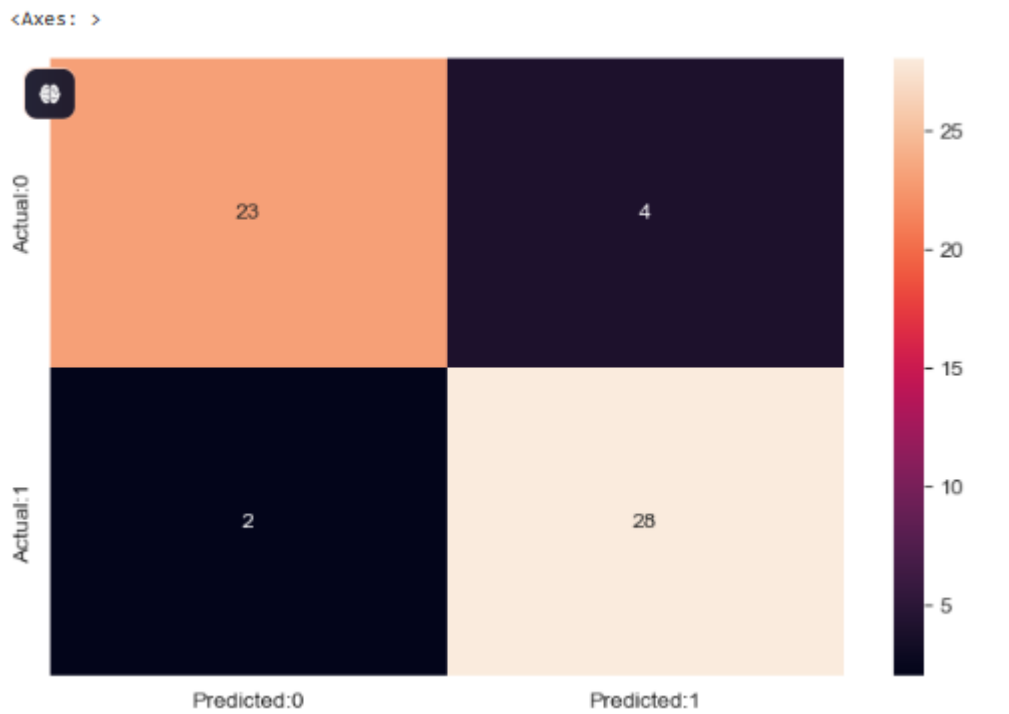


Figure 4.1.5 Confusion Matrix Analysis

4.1.6 SHAP-Based Feature Analysis

To improve the interpretability of the proposed Heart Attack Risk Prediction System, SHAP (SHapley Additive Explanations) was applied to analyze the contribution of individual features toward the prediction outcome. SHAP provides a clear explanation of how each patient attribute influences the model's decision, making the prediction process transparent and easier to understand.

The SHAP analysis revealed that **maximum heart rate achieved (thalachh)** was the most influential feature, with a contribution value of approximately **0.604**, indicating a significant impact on heart disease prediction. Other important features included **number of major vessels (caa)** with a contribution of **0.492**, **sex** with **0.444**, and **exercise-induced angina (exng)** with **0.285**. Additional contributing features were **thal**, **slope of the ST segment (slp)**, **resting ECG results (restecg)**, **resting blood pressure (trtbps)**, and **oldpeak**, all of which played a role in determining the patient's risk level.

The analysis also identified the factors that increased heart disease risk. Features such as **thalachh**, **caa**, **exng**, **thal**, **slp**, **restecg**, **trtbps**, and **oldpeak** showed positive contributions toward the prediction, indicating a higher likelihood of heart disease. In contrast, features with lower contribution values had a smaller influence on the final prediction outcome.



Figure 4.1.6 SHAP-Based Feature Analysis

4.1.7 New Patient Prediction Results

To validate the effectiveness of the proposed Heart Attack Risk Prediction System, the trained machine learning model was tested using new patient records that were not included in the training dataset. The purpose of this evaluation was to determine the model's ability to generalize to unseen data and provide accurate predictions in real-world healthcare scenarios. The system accepts patient information such as age, sex, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, ST depression (oldpeak), slope of the ST segment, number of major vessels, and thalassemia status.

After receiving the patient data, the model performs preprocessing and feeds the information into the selected machine learning classifier. The system then generates a prediction indicating whether the patient has a **low risk (0)** or **high risk (1)** of heart disease. In addition to the classification result, the model also provides a probability score that reflects the confidence level of the prediction. This probability score helps healthcare professionals assess the certainty

of the prediction and make informed decisions regarding further medical evaluation or treatment.

For example, when a new patient's health information is entered into the system, the model analyzes the relationship between the patient's attributes and the patterns learned during training. If the patient's characteristics closely resemble those of previously identified high-risk patients, the system predicts a high risk of heart disease. Conversely, if the patient's attributes indicate healthier cardiovascular conditions, the model predicts a low risk. The prediction result is displayed instantly, enabling rapid risk assessment and early intervention.

The screenshot displays a web application for heart attack prediction. It features a form with 16 input fields for patient data, a 'Predict Heart Risk' button, a success message, and a large red alert box indicating a high risk. Below the alert is a 'Risk Meter' visualizing the 91.65% probability.

Attribute	Value
age	57
sex	1
cp	1
trtbps	120
chol	236
fbs	0
restecg	1
thalachth	178
exng	0
oldpeak	0.8
slp	2
caa	0
thall	2

Predict Heart Risk

📄 Patient Data Saved Successfully

🚨 HIGH HEART ATTACK RISK
Risk Probability: 91.65%

Risk Meter

91.65%

Figure 4.1.7 New Patient Prediction Results

4.2 Snapshots of Results

This section presents the snapshots of the developed system interface used in the proposed Heart Attack prediction project. The snapshots includes

age	37
sex	1
cp	2
resttps	130
chol	204
fbx	0
restecg	0
thalachh	172
exng	0
oldpeak	1.4
slp	2
caa	0
thall	2

Predict Heart Risk

Figure 4.2.1 Patient Data Input Interface

The Figure 4.2.1 illustrates the user input interface of the proposed heart attack risk prediction system. In this interface, users are required to enter values for all 13 features used in the dataset, including demographic, clinical, and lifestyle-related attributes. These input parameters serve as the basis for the machine learning model to assess the patient's heart attack risk. The interface ensures that all necessary information is collected before prediction, enabling the model to generate accurate and reliable risk assessments. The entered data is subsequently processed by the trained model and stored for future analysis and record maintenance.

The screenshot displays a web application interface for heart attack prediction. It features a list of 13 input fields for patient data, a 'Predict Heart Risk' button, a success message, a high-risk alert, and a risk meter.

Feature	Value
age	57
sex	1
cp	1
trtbps	120
chol	236
fbs	0
restecg	1
thalachh	178
exng	0
oldpeak	0.8
slp	2
caa	0
thall	2

Predict Heart Risk

📄 Patient Data Saved Successfully

🚨 HIGH HEART ATTACK RISK
Risk Probability: 91.65%

Risk Meter

91.65%

Figure 4.2.2 High-Risk Prediction Result

Figure 4.2.2 illustrates a patient case that has been classified as **High Risk** by the proposed heart attack prediction system. Based on the values entered for the 13 health-related features, the trained machine learning model predicts a high likelihood of heart attack occurrence. The system displays the probability associated with the high-risk classification along with the patient's feature values.

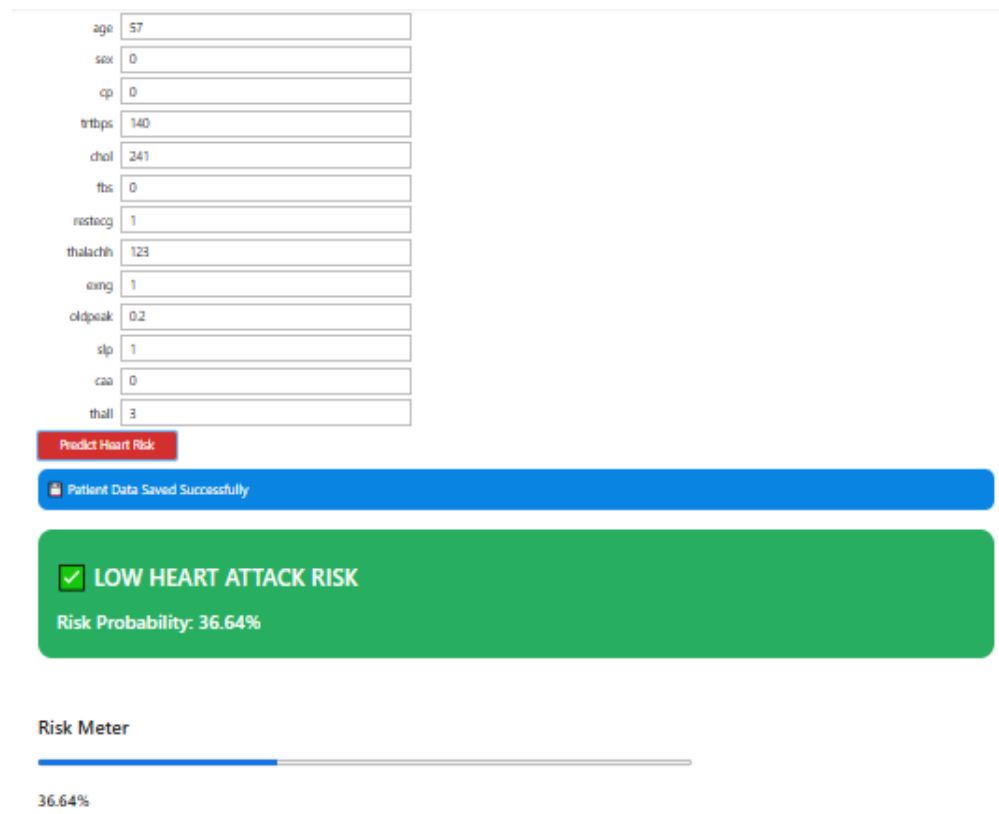


Figure 4.2.3 Low-Risk Prediction Result

Figure 4.2.3 represents a patient case that has been classified as **Low Risk** by the proposed system. After analyzing the input values of all 13 features, the model determines that the likelihood of a heart attack is comparatively low. The prediction output includes the probability score for the low-risk category and displays the corresponding feature values used in the assessment. This result suggests that the patient's health indicators are generally within safer ranges, reducing the immediate risk of cardiovascular complications according to the model's evaluation.



Figure 4.2.4 XAI analysis

Figure 4.2.4 represents XAI helps identify the contribution of individual features such as age, cholesterol level, blood pressure, diabetes status, smoking habits, and physical activity toward the predicted heart attack risk. By quantifying the impact of each feature, the system enables users and healthcare professionals to understand why a patient is classified as high risk or low risk. Features that increase the predicted risk are shown as positive contributors, while features that reduce the risk are shown as negative contributors.

Furthermore, XAI improves trust and reliability in the prediction process by making model decisions transparent and understandable. It assists medical practitioners in validating the model's recommendations and supports informed clinical decision-making. The feature importance analysis generated through SHAP also helps identify the most influential risk factors across the entire dataset, providing valuable insights into cardiovascular health patterns.

age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	Prediction	Risk_Probability	Timestamp
40	0	2	110	200	0	1	123	1	0.3	1	0	3	High Risk	97.34	58:10.4
65	1	3	160	280	1	2	105	1	3.5	2	3	3	Low Risk	1.28	03:21.9
35	0	0	118	180	0	0	170	0	0.1	1	0	1	High Risk	97.34	04:42.1
57	0	0	120	354	0	1	163	1	0.6	2	0	2	High Risk	85.62	12:44.2

Figure 4.2.6 New Patient Data Storage

Figure 4.2.6 represents The proposed heart attack risk prediction system includes a data storage mechanism that records the details of newly entered patients along with their prediction results. After the user provides values for all 13 input features and the model generates a prediction, the patient information is automatically stored in a separate data file or database for future reference. This stored information includes the patient's health attributes, predicted risk category (High Risk or Low Risk), prediction probability, and the date and time of prediction. The data storage functionality enables efficient management of patient records and allows previously generated predictions to be reviewed whenever required. It also supports future analysis, model validation, and the continuous expansion of the dataset with new patient records. By maintaining a structured repository of patient information and prediction outcomes, the system ensures traceability, facilitates record keeping, and enhances the practical usability of the heart attack risk prediction framework in real-world healthcare environments.

4.3 Summary

This chapter presented the experimental results and analysis of the proposed heart attack risk prediction system. The study began with data preprocessing and exploratory data analysis to understand the characteristics of the dataset and identify important patterns among the health-related features. Correlation analysis was performed to examine the relationships between the input variables and the target outcome, helping to identify significant risk factors associated with heart attacks.

Several machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), AdaBoost, Gradient Boosting, and XGBoost, were trained and evaluated. Their performance was compared using various evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score. The comparative analysis enabled the selection of the best-performing model for heart attack risk prediction.

The chapter also presented confusion matrix and ROC curve analyses to assess classification effectiveness. To improve transparency and interpretability, SHAP-based Explainable Artificial Intelligence (XAI) techniques were applied, allowing the contribution of each feature to be clearly understood. Both global feature importance and individual patient-level explanations were analyzed to provide insights into model decisions.

Furthermore, prediction results for new patient data were demonstrated through high-risk and low-risk case studies, along with probability scores and feature contributions. The system successfully stored patient records and prediction outcomes for future reference. Overall, the experimental findings confirm that the proposed framework provides accurate, reliable, and explainable heart attack risk predictions, making it a valuable tool for supporting early cardiovascular risk assessment and healthcare decision-making.

5. Conclusion and future scope

5.1 Conclusion

This project successfully developed an intelligent and explainable heart attack risk prediction system using machine learning and SHAP-based Explainable Artificial Intelligence (XAI). The study addressed the challenge of accurately identifying individuals at risk of heart attacks by analyzing patient health data through a structured process involving data preprocessing, exploratory data analysis, correlation analysis, model development, and performance evaluation.

Multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), AdaBoost, Gradient Boosting, and XGBoost, were implemented and compared. Experimental results demonstrated that the selected best-performing model achieved strong predictive performance in classifying patients into high-risk and low-risk categories. The evaluation metrics confirmed the effectiveness and reliability of the proposed framework for heart attack risk prediction.

A key contribution of this work is the integration of SHAP-based XAI, which provides clear explanations of how individual features influence prediction outcomes. This enhances model transparency, improves user trust, and enables healthcare professionals to better understand the factors contributing to patient risk. The system also supports new patient data entry, probability-based risk prediction, feature contribution analysis, and storage of prediction records for future reference.

Overall, the project successfully achieved its objectives by delivering an accurate, reliable, and interpretable heart attack risk prediction system. The combination of machine learning, explainability, and patient data management demonstrates the practical value of the proposed approach in supporting early cardiovascular risk assessment and informed healthcare decision-making.

5.2 Future Scope

The proposed heart attack risk prediction system provides a strong foundation for intelligent cardiovascular risk assessment; however, several enhancements can be implemented to further improve its effectiveness and practical applicability. Future work can focus on expanding the dataset by incorporating patient records from multiple healthcare institutions, which would

improve the model's robustness, generalization capability, and reliability across diverse populations.

Additional clinical parameters such as ECG measurements, blood test reports, family medical history, medication records, and medical imaging data can be integrated into the system to provide a more comprehensive evaluation of heart attack risk. The inclusion of such data may further enhance prediction accuracy and enable the identification of complex health patterns that are not captured by the current feature set.

The system can also be deployed as a web-based or mobile healthcare application, allowing patients and healthcare professionals to access risk predictions anytime and anywhere. Integration with Electronic Health Records (EHRs), wearable devices, and IoT-based health monitoring systems would enable automatic collection of real-time patient data and continuous health monitoring. This would support proactive healthcare management and early intervention for high-risk individuals.

Furthermore, future work can include real-world clinical validation through collaboration with hospitals and healthcare organizations. Such validation would assess the effectiveness of the system in practical medical environments and support its adoption as a clinical decision-support tool. With these advancements, the proposed framework has the potential to evolve into a comprehensive, intelligent, and explainable healthcare platform that supports accurate risk prediction, continuous patient monitoring, and improved cardiovascular disease prevention.

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